Thunkable: Blocks for Events, Variables & Logic

Make your app do things when users tap, type and choose.

Thunkable | Beginner to Intermediate | Ages 10–16

InfiBee Coders

💡 The Big Idea

- Blocks are the instructions of your app. You snap them together like puzzle pieces.
- An event block waits for something, like a tap, and then runs the blocks inside it.
- Variables store values like scores or names.
- Logic blocks, like if and else, let your app make choices.
- Block names can look slightly different as Thunkable updates, so use the drawers to find them.

⚡ Events and Actions

- `when Button1.Click do` runs when the user taps the button.
- `set Label1.Text to` changes the words on a label.
- `TextInput1.Text` gives you what the user typed.
- `open screen` (Control drawer) moves to another screen.

📦 Variables and Text

- Create a variable in the Variables drawer and give it a clear name, like `count`.
- Set it to a starting value, like 0.
- Update it with `set` and a maths block, like `count + 1`.
- Use `join` to put text and numbers together in one message.

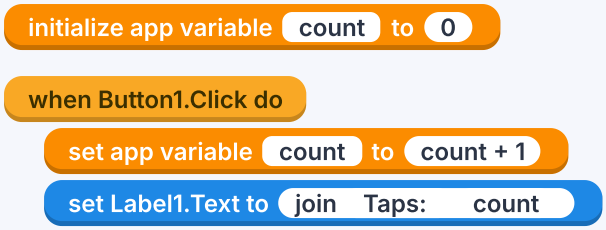
🔀 Logic

- `if` runs blocks only when a condition is true.
- `else` runs when the condition is false.
- Comparison blocks include `=`, `<` and `>`.
- `and`, `or` and `not` combine conditions.

💻 Block Examples

Tap counter

Thunkable blocks

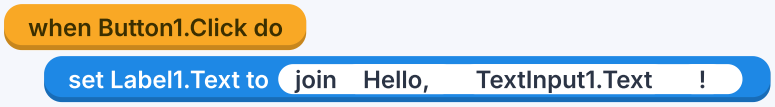


▶ What happens:

Every tap adds 1, and the label shows Taps: 1, Taps: 2 and so on.

Greeting app

Thunkable blocks

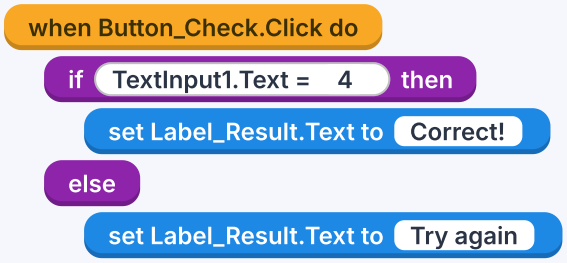


▶ What happens:

If the user types Mia and taps the button, the label shows Hello, Mia!

Quiz answer check

Thunkable blocks

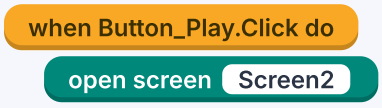


▶ What happens:

Typing 4 shows Correct!. Anything else shows Try again.

Open another screen

Thunkable blocks



▶ What happens:

Tapping Play takes the user to Screen2.

📚 Key Terms

Block
One instruction you snap into place.

Condition
A true or false test.

Event
Something the user does that starts blocks.

If / else
A choice between two paths.

Click
A tap on a button.

Join
A block that glues text together.

Variable
A named box for a value.

Property
A setting of a component, like its Text.

Initialize
Give a variable its first value.

Bug
A mistake that makes the app act wrongly.

🧮 Quick Reference

```
when Button1.Click do ...
```

```
set Label1.Text to ...
```

```
TextInput1.Text
```

```
initialize app variable [count] to (0)
```

```
join [Hi ] (name)
```

```
if <a = b> then ... else ...
```

```
open screen [Screen2]
```

✏️ Worked Examples

Q: Make a button that changes a label to Hello!

A: Step 1: Add Label1 and Button1. Step 2: In Blocks, add when Button1.Click do. Step 3: Inside, add set Label1.Text to [Hello!].

Q: Count how many times a button is tapped.

A: Step 1: Create a variable count set to 0. Step 2: In the click event, set count to count + 1. Step 3: Show it with set Label1.Text to count.

Q: Show Correct! only if the user types the right answer.

A: Use an if block with TextInput1.Text = the right answer, set the label to Correct!, and add an else for Try again.

🚫 Common Mistakes

- Forgetting to set the variable's starting value.

- Changing a label's text in the Design view when it should change in Blocks.
- Putting blocks outside the event block, so they never run.
- Comparing text with the wrong spelling or capital letters.
- Using `set` when you meant to change by adding to the old value.

👨‍👩‍👧 Parent Corner

- Ask your child to explain in words what a block stack does.
- Let them predict the result before pressing Test.
- Encourage small changes, like different greeting words.
- Talk about how real apps use the same idea: tap, then react.
- Celebrate the first time an app responds to a tap.

🎯 Quick Check

1. Which block runs when a button is tapped?
✅ when Button1.Click do.

2. How do you read what the user typed?
✅ TextInput1.Text.

3. What does a variable do?
✅ It stores a value that can change.

4. What does else do in an if block?
✅ It runs when the condition is false.

5. What does join do?
✅ It glues text and values together into one message.